

HAT-P-32b

R-0577

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_221230 · created 2026-07-24T17:22:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459943.8232 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.053
Transit depth (Rp/Rs)^2	2.93 +/- 1.81 [%]
Orbital Inclination (inc)	85.71 +/- 1.74
Airmass coefficient 1 (a1)	4.3156 +/- 0.0088
Airmass coefficient 2 (a2)	-0.8 +/- 0.16
Scatter in the residuals of the lightcurve fit is	0.3 %
Best Comparison Star	#1 - [454, 252]
Optimal Aperture	24.17
Optimal Annulus	112.77
Transit Duration (day)	0.1264 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.053 vs 0.1489 (deviation 0.42σ)
Duration fit vs published	0.1264 d vs 0.1304 d (deviation 0.5σ)
Mid-time O–C	7.5 min
Depth SNR	3.2
Residual scatter	0.3 %

Light curve

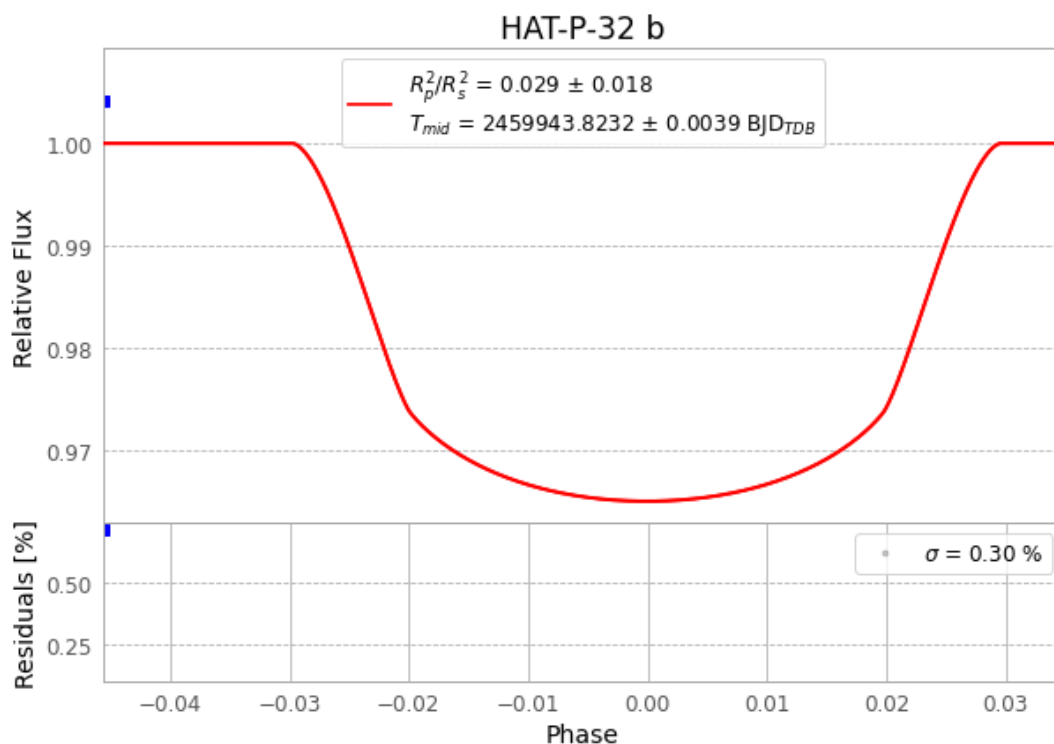


Figure 1 — FinalLightCurve_HAT-P-32 b_2022-12-29.png (SHA-256 b5633fd35a1feb97...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [177, 27], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 34230246f2941a47...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

88 FITS frames (58 MB), HATP-32221230050015.FITS → HATP-32221230092715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-221230.log (1e5c700a6b39...), FinalLightCurve_HAT-P-32 b_2022-12-29.png (b5633fd35a1f...), FinalLightCurve_HAT-P-32 b_2022-12-29.csv (b2ce000b56c7...)

Compute resources

Wall time 746.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 17:22Z.